DATASCI 151 - Introduction to Statistical Computing II

Danilo Freire

Summer 2026

Contents

Course Description 2

Learning Objectives 2

Prerequisites 2

Materials 3
- Suggested Books 3
- Online Courses 3
- Additional Resources 3

Course Information 4

Software 4

Office Hours 4

Academic Integrity 5

Artificial Intelligence 5

Special Needs and Accessibility Services 5

English Language Learners 6

Assignments and Grading Policy 6

Grading Scale 6

Course Outline and Suggested Readings 7

Course Description

Welcome to DATASCI 151! This Maymester course introduces data analysis and statistical computing using Python and SQL. It is perfect for beginners with little or no programming experience who want to build skills for data-driven decision making.

We will cover Python programming fundamentals, Jupyter Notebooks for reproducible research, data wrangling and merging with SQL, data visualisation, and introductions to linear modelling and time series analysis.

You will work with real-world datasets and problems, gaining hands-on experience using these tools to extract insights from data. The course develops both technical skills and critical thinking abilities needed for tackling complex data challenges.

Learning Objectives

By the end of this course, you will be able to:

1. Write Python functions and perform mathematical operations
2. Wrangle and manipulate data using Python libraries like Pandas
3. Merge and manage databases with SQL
4. Create visualisations to communicate data insights
5. Implement linear models and understand time series analysis principles
6. Use Jupyter Notebooks for reproducible research
7. Develop problem-solving skills for data analysis and statistical computing

Prerequisites

There are no prerequisites for this course. Everyone is welcome, regardless of prior programming or data analysis experience. Feel free to reach out if you have questions about the course content or whether it is right for you.

Materials

This course provides everything you need to master the core concepts, but you are encouraged to explore these additional resources to deepen your understanding of Python and SQL.

Suggested Books

- Python for Data Analysis by Wes McKinney
- Elements of Data Science by Allen Downey
- Automate the Boring Stuff with Python by Al Sweigart
- Python for Everybody by Charles Severance
- SQL for Data Scientists by Renee M. P. Teate

Online Courses

- Coursera: Python for Everybody Specialisation
- edX: Python Basics for Data Science
- Codecademy: Learn Python
- DataCamp: Introduction to SQL
- Coursera: SQL for Data Science

Additional Resources

- Python Documentation
- Pandas Documentation
- SQLite Documentation
- SQLBolt
- DataLemur for SQL interview practice
- Github Guides

Course Information

We will meet online Monday through Friday from 11:00am to 12:20pm EDT. Please review the materials before each class. All course information is available on our GitHub repository at `https://github.com/danilofreire/datasci151-summer`. While I intend to follow the schedule as planned, I also want to adapt to your learning pace and style, so the syllabus may change during the course. Please check the course repository regularly for updates, which I will also announce in class and on Canvas.

Software

We will mainly use Python in this course, which is a free and general-purpose programming language widely used in data science and machine learning. I recommend the Anaconda distribution, which already includes many useful libraries like Pandas, NumPy, and Jupyter.

You can use any text editor, but I recommend VS Code with the Python extension. PyCharm is another excellent option. Feeling adventurous? Try Neovim with the coc-pyright plugin—that is, if you can exit the editor. :)

We will also use Jupyter Notebooks in class. Jupyter comes pre-installed with Anaconda, but please install the Jupyter extension for VS Code too. We will have a hands-on session to use Jupyter effectively.

I have prepared a series of tutorials to help you install Anaconda, Jupyter, VS Code, and set up a free GitHub educational account. Please work through these tutorials as soon as possible to ensure you have all the necessary tools.

Office Hours

I am flexible with office hours, but email works best. Feel free to message me anytime at `danilo.freire@emory.edu`. I will usually reply within a few hours.

Academic Integrity

Every Emory University community member is responsible for upholding a standard of unimpeachable honour in all academic work. The Honour Code of Emory College assumes that every member of the University will not only conduct their own life according to the highest ethical standards but will also refuse to tolerate actions in others that would harm the institution's reputation. Academic misconduct includes any action or inaction that violates the integrity and honesty of the academic community. Any suspected cases will be referred to the Emory Honour Council.

Artificial Intelligence

You will complete five problem sets and three in-class quizzes. You may use AI tools for assignments as a supplement to your own thinking, not a replacement for it. **AI tools are not permitted during quizzes**, which are individual tests of your understanding. If you use AI on an assignment, check the output carefully and take responsibility for any errors in what you submit. Cite all sources, including AI tools, and add a brief note at the end of any document where AI was used.

Special Needs and Accessibility Services

I am committed to providing accommodations so all students can succeed. If you have medical or health conditions that might affect your performance, please visit the Department of Accessibility Services (DAS) to learn about available accommodations. Provide me with your DAS Accommodation Letter at the start of the semester or when it is granted.

Note: DAS accommodations like extra time apply only to quizzes, not assignments (since assignments are released in advance, letting you work at your own pace). Athletes and students with other commitments should inform me of scheduling conflicts early in the course. I will do my best to accommodate you, though I cannot guarantee every request. Questions? Just reach out.

English Language Learners

Emory University welcomes students from across the country and around the world. The unique perspectives that international and multilingual students bring enrich our entire community. To support multilingual learners, we offer language and culture workshops and individual appointments. For more information about English Language Learning support, contact the ELLP Specialists at `https://writingcenter.emory.edu`. Your English proficiency will never be penalized.

Assignments and Grading Policy

Problem Sets (50%). You will complete five problem sets throughout the course. These reinforce lecture concepts and give you hands-on practise with statistical programming. Each set includes theoretical questions and practical applications that you must complete individually. Acknowledge all sources, including textbooks, articles, and AI tools.

Late submissions receive half credit automatically. Please submit all assignments as HTML files, Jupyter notebooks (`.ipynb`), or PDFs via Canvas by midnight on the due date.

Class Quizzes (50%). You will take three in-class quizzes on Fridays, based on that week's lectures. Quizzes test your understanding and ability to apply concepts to new problems. They are open-book and open-notes, but **AI tools are not allowed**. Quizzes are individual assessments—no discussing questions with classmates.

Grading Scale

Each student's final grade will be based on the following percentage ranges, rounded up to the nearest point:

Grade	A	A-	B+	B	B-	C	D	F
Range	91–100	86–90	81–85	76–80	71–75	66–70	60–65	<60

Course Outline and Suggested Readings

The lecture notes cover all essential material we will cover. As mentioned, the course outline may change, so please check our GitHub repository regularly for updates.

Each lecture folder contains an HTML file or a Jupyter notebook (`.ipynb`) with code examples and explanations, along with any additional resources or datasets used in the lecture.

- Tuesday, May 12: Lecture 01: Welcome to DATASCI 151 - Introduction and Course Overview. Please make sure to install the necessary software for the course by following the Course Tutorials: How to Install Anaconda, Jupyter, VSCode, and Open a Free Educational Account on GitHub.
- Wednesday, May 13: Lecture 02: Jupyter Notebooks, Packages, Variables, and Lists. Lecture Slides, Lecture Notebook.
- Thursday, May 14: Lecture 03: Maths Operations, Arrays, and Boolean Logic. Lecture Slides, Lecture Notebook
  - Kahoot Quiz. Kahoot quizzes are optional and will not be graded. We will do it in class! It is a fun way to review the material covered in the lecture.
- Friday, May 15: Lecture 04: Quiz 01 - Based on Lectures 01-03. The repository contains the quiz questions and will be made online on the day of the quiz.
  - Assignment 01 due.
- Monday, May 18: Lecture 05: Conditional Statements. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.
- Tuesday, May 19: Lecture 06: While and For Loops. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.
- Wednesday, May 20: Lecture 07: Functions. Lecture Slides, Lecture Notebook.

  - Kahoot Quiz.
  - Assignment 02 due.

- Thursday, May 21: Lecture 08: Data Wrangling with Pandas. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.

- Friday, May 22: Lecture 09: Quiz 02 - Based on Lectures 05-08. As with Quiz 01, the repository will be available on the day of the quiz.
  - Assignment 03 due.

- Monday, May 25: Lecture 10: Subsetting and Filtering Data. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.

- Tuesday, May 26: Lecture 11: Introduction to SQL. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.

- Wednesday, May 27: Lecture 12: SQL in Python. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.
  - Assignment 04 due.

- Thursday, May 28: Lecture 13: Merge Tables in SQL. Lecture Slides, Lecture Notebook.
  - Kahoot Quiz.

- Friday, May 29: Lecture 14: Quiz 03 - Based on Lectures 10-13.
  - Assignment 05 due.